CHENNAI RAMAPURAM
SRM
INSTITUTE OF SCIENCE & TECHNOLOGY
(Deemed to be University u/s 3 of UGC Act, 1956)

DEVELOPER STUDENTS CLUB
SRM IST RMP

TITLE

Fintech

UPI Secure Pay AI

"Blocks UPI fraud before money leaves the account, not after."

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HACKCELERATE

SENTINELS SQUAD

2nd year CSE Students, Rajalakshmi Engineering College.

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PROBLEM STATEMENT

1.4 Lakh Crore Lost: Why UPI Fraud Happens in Seconds

India's UPI processes 20-22 billion transactions monthly across 500M users, powering 85% of retail digital payments. From the FY **2020-21 to 2024-25**, the total amount lost due to UPI-related fraud in India was approximately **₹3,883 crore, a** number that's accelerating rapidly.

Why Current UPI Fraud Systems Fail

- Only 70% detection rate (industry average)
- 15% false positives rate (innocent users blocked)
- Mostly rely on rule-base checks
- Limited behavioral analysis (only check amounts)

Real-World Impact (Based on reported case 2024)

- Elderly users lose life savings to SIM swap scams
- Small merchants face 2-3 fraud attempts monthly
- Maximum of UPI fraud victims are over 50 years old*

Source: RBI Annual Reports, Ministry of Finance Parliamentary Replies (FY 2021–2025)

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SOLUTION

UPI Secure Pay is a real-time AI system that prevents UPI payment frauds before money is transferred.

It is demonstrated as a prototype using simulated UPI transaction flows and device-level signals.

it acts as a smart bodyguard during every UPI transaction by continuously monitoring user behavior and transaction patterns. The system uses a **four-layer fraud detection** approach to analyze behavioral signals, detect anomalies, track suspicious money flows, and apply safety rules to ensure secure and seamless digital payments.

Designed to support sub-second decisions using lightweight models and parallel execution in a prototype environment". Transactions are approved, blocked, or flagged in real time to prevent financial loss.

01 | **Behavioral Analysis** : Learns how users normally swipe, hold their phone, device fingerprints, App usages pattern.

02 | **Anomaly Detection** : Identifies unusual transactions such as sudden high-value payments, odd timing, or unknown merchants

03 | **Money Trail Tracking** : Used to update risk scores of recipients and networks for future transactions.

04 | **Rule-Based Safety Layer** : Instantly blocks known fraud patterns like multiple wrong PIN attempts on a new device or suspicious amounts. Example: New device + high amount → step-up verification

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CHALLENGES & MITIGATIONS

1. **False positives blocking genuine users** – Mitigated using user-specific baseline thresholds and gradual risk scoring instead of binary blocking
2. **Limited access to real fraud data** – Synthetic datasets are generated to mirror RBI-reported transaction distributions and validated against known fraud patterns.
3. **Evolving and unseen fraud patterns** – Anomaly detection is used as an early warning layer, with human-review or rule updates for confirmed cases.
4. **Strict real-time performance requirements** – “Models are optimized for sub-200 ms inference on CPU-only environments.
5. **User privacy and data security risks** – No raw biometric or PIN data is stored; only hashed device and session-level features are processed in memory.

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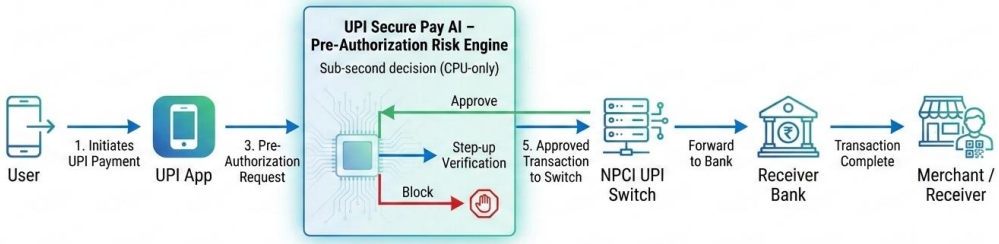
ADDITIONAL INFO

UPI Secure Pay AI – Pre-Authorization Integration in UPI Transaction Flow



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TECH STACK

React & Tailwind CSS – Interactive and visually appealing dashboard

WebSocket & Recharts – Real-time transaction updates and data visualizations

FastAPI – High-performance Python backend for fraud detection

XGBoost, Isolation Forest & NetworkX – transaction risk scoring, unknown pattern detection, and offline graph analysis.

PostgreSQL – Fraud records are stored in PostgreSQL for audit and analysis

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TEAM MEMBERS

Team Name

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